

Sample Paper 4 — Class 5 Mathematics Olympiad

Pattern: SOF IMO Level 1 — 50 questions, 60 marks, 60 minutes, no negative marking. Set a timer before you start.

Section 1 — Logical Reasoning (Q1–Q15, 1 mark each)

- Q1.** Find the odd one out: A) Rose B) Lily C) Lotus D) Carrot
- Q2.** Key is to Lock as Password is to _____. A) Computer B) Account C) Door D) Car
- Q3.** If each letter of a word is moved 5 steps forward in the alphabet, BAG becomes GFL. Using the same rule, how is CAP written? A) HFU B) HFV C) GFU D) HGU
- Q4.** Complete the series: 9, 18, 36, 72, ____ A) 108 B) 126 C) 144 D) 162
- Q5.** In a row of 28 students, Tarun is 16th from the left. What is his position from the right? A) 12th B) 13th C) 14th D) 11th
- Q6.** Facing West, Sahil turns 90° clockwise, then another 90° clockwise. Which direction does he face now? A) North B) South C) East D) West
- Q7.** Arrange in a logical order of a butterfly's life cycle: 1. Egg 2. Caterpillar 3. Butterfly 4. Cocoon
A) 1, 2, 4, 3 B) 1, 2, 3, 4 C) 1, 4, 2, 3 D) 2, 1, 4, 3
- Q8.** Find the odd one out: A) 14 B) 21 C) 25 D) 35
- Q9.** If today is Sunday, what day will it be after 40 days? A) Thursday B) Friday C) Saturday D) Monday
- Q10.** Pointing to a man, Priya said, "He is the husband of my mother's sister." How is the man related to Priya? A) Father B) Uncle C) Brother D) Cousin
- Q11.** Find the missing number: 5, 11, 23, 47, ____ A) 91 B) 93 C) 95 D) 97
- Q12.** In a certain code, STONE is written as UVQPG (each letter moved 2 steps forward). How is PLANT written in the same code? A) RNCPU B) RNCPU C) RMCPV D) SNCPV
- Q13.** Find the odd one out: A) Circle B) Oval C) Sphere D) Square
- Q14.** X is heavier than Y. Z is lighter than Y. W is heavier than X. Who is the heaviest? A) X B) Y C) Z D) W
- Q15.** Complete the series: 2, 3, 5, 8, 13, ____ A) 18 B) 19 C) 20 D) 21

Section 2 — Mathematical Reasoning (Q16–Q35, 1 mark each)

- Q16.** $78,456 + 16,789 = ?$ A) 95,245 B) 95,145 C) 94,245 D) 95,345
- Q17.** $60,004 - 27,658 = ?$ A) 32,346 B) 32,246 C) 32,446 D) 33,346
- Q18.** $3,789 \times 8 = ?$ A) 30,312 B) 30,212 C) 30,412 D) 29,312
- Q19.** $68,432 \div 8 = ?$ A) 8,554 B) 8,544 C) 8,564 D) 8,654
- Q20.** Find the HCF of 40 and 60. A) 10 B) 15 C) 20 D) 30
- Q21.** Find the LCM of 10 and 15. A) 30 B) 60 C) 15 D) 150
- Q22.** Which fraction is equivalent to $\frac{7}{10}$? A) $\frac{14}{20}$ B) $\frac{10}{7}$ C) $\frac{7}{20}$ D) $\frac{14}{10}$
- Q23.** $\frac{4}{9} + \frac{2}{9} + \frac{1}{9} = ?$ A) $\frac{7}{9}$ B) $\frac{7}{27}$ C) $\frac{6}{9}$ D) 1
- Q24.** Convert 0.32 to a fraction in simplest form. A) $\frac{8}{25}$ B) $\frac{4}{25}$ C) $\frac{8}{5}$ D) $\frac{16}{25}$
- Q25.** What is 40% of 150? A) 50 B) 55 C) 60 D) 65
- Q26.** Express $\frac{11}{20}$ as a percentage. A) 50% B) 55% C) 60% D) 65%
- Q27.** Two numbers are in the ratio 7:9. If the smaller is 35, find the larger. A) 40 B) 42 C) 45 D) 48
- Q28.** The cost price of an item is ₹600 and the selling price is ₹750. Find the profit percentage. A) 20% B) 25% C) 30% D) 35%
- Q29.** Find the simple interest on ₹7,000 at 4% per annum for 6 years. A) ₹1,580 B) ₹1,620 C) ₹1,680 D) ₹1,720
- Q30.** Convert 6 kg 45 g into grams. A) 6,045 B) 6,450 C) 6,405 D) 645
- Q31.** Convert 225 minutes into hours and minutes. A) 3 h 25 min B) 3 h 35 min C) 3 h 45 min D) 3 h 55 min
- Q32.** Find the perimeter of a rectangle with length 31 cm and breadth 17 cm. A) 48 cm B) 96 cm C) 192 cm D) 527 cm
- Q33.** Find the area of a square with side 13 cm. A) 156 sq cm B) 169 sq cm C) 26 sq cm D) 179 sq cm
- Q34.** How many lines of symmetry does a circle have? A) 1 B) 2 C) 4 D) Countless (infinite)
- Q35.** How many right angles make two full turns? A) 4 B) 6 C) 8 D) 10

Section 3 — Everyday Mathematics (Q36–Q45, 1 mark each)

- Q36.** A factory makes 2,750 pens each day. How many pens does it make in 40 days? A) 1,00,000 B) 1,05,000 C) 1,10,000 D) 1,15,000
- Q37.** Neha had ₹7,200. She spent ₹2,450 and then ₹1,890. How much money does she have left? A) ₹2,760 B) ₹2,860 C) ₹2,960 D) ₹2,660

- Q38.** A tank holds 960 L. If $\frac{5}{8}$ of it is used, how much water is left? A) 340 L B) 350 L C) 360 L D) 370 L
- Q39.** 32 kg of rice costs ₹1,600. Find the cost of 1 kg. A) ₹45 B) ₹48 C) ₹50 D) ₹52
- Q40.** 14 rows of chairs have 23 chairs each. How many chairs are there in total? A) 312 B) 322 C) 332 D) 342
- Q41.** A man deposits ₹4,500 at 8% simple interest per year. How much interest will he earn in 5 years? A) ₹1,700 B) ₹1,750 C) ₹1,800 D) ₹1,850
- Q42.** A trader bought goods for ₹2,100 and sold them for ₹2,415. Find the profit percentage. A) 10% B) 15% C) 20% D) 25%
- Q43.** A cyclist rides at 18 km/h. How far will they travel in 2 hours 50 minutes? A) 45 km B) 48 km C) 51 km D) 54 km
- Q44.** A recipe needs $\frac{7}{8}$ cup of oil. How much oil is needed for 8 such recipes? A) 6 cups B) 7 cups C) 8 cups D) 9 cups
- Q45.** In a survey of 400 students, 42% like swimming. How many students like swimming? A) 158 B) 163 C) 168 D) 173

Section 4 — Achievers Section (Q46–Q50, 3 marks each)

- Q46.** A number is decreased by 35% to get 390. Find the original number. A) 550 B) 580 C) 600 D) 620
- Q47.** The sum of four consecutive numbers is 106. Find the largest number. A) 26 B) 27 C) 28 D) 29
- Q48.** The simple interest on a sum for 6 years at 5% per annum is ₹2,700. Find the sum. A) ₹8,000 B) ₹8,500 C) ₹9,000 D) ₹9,500
- Q49.** Ananya had some money. She spent $\frac{3}{8}$ of it on clothes and $\frac{1}{5}$ of the remainder on books. She is left with ₹1,000. How much money did she have at first? A) ₹1,800 B) ₹2,000 C) ₹2,200 D) ₹2,400
- Q50.** Two numbers are in the ratio 5:7. If each number is increased by 8, the new ratio becomes 3:4. Find the smaller of the two original numbers. A) 35 B) 40 C) 45 D) 50

Answer key: `sample-paper-04-answer-key.md`
